



Meet Python Turtle

Quick facilitation notes & answer key

Goal

Students meet typed Python by reading a for-loop that repeats a forward move — bridging the repeat block they know to `range(n)` in code (the block→text bridge). First typed code; angles/turns come next sheet.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle projected (or a Turtle web playground); the printed worksheet + a pencil per student. Optional: a device per pair to run the code.

Run it (about 30 min)

1. Recall the repeat block from Block Coding — show it beside the Python for-loop.
2. Read `range(3)` together as "do it 3 times"; run `forward(20)` and count the hops.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: `range(n)` = repeat `n`, just typed.

Answer key (checked by the run-gate)

Ex 1 — 3 hops; 60 steps total.

Ex 2 — 5 hops; 100 steps; predicted = 100.

Ex 3 — `range(4)`, `forward(10)`.

Ex 4 (challenge) — 100 steps (4×25).

Watch for & differentiate

Common slip: reading `range(3)` as 3 + the start (thinking 4 hops) — it is exactly 3.

Support: point at the for-line and say "do the indented line this many times".

Extend: ask total steps for `range(6)`, `forward(10)` (60).

Success looks like

The student reads `range(n)` as "repeat `n` times", counts the hops and total distance (Ex 1–2), completes a for-loop line (Ex 3), and works out a total from a typed loop (Ex 4).